

Summary of A Survey on Mixture of Experts

A Survey on Mixture of Experts: Summary

Overview and Motivation

This survey provides the first comprehensive review of Mixture of Experts (MoE) models since 2022, addressing a critical gap in the literature following the rapid advancements after the "ChatGPT moment." As large language models (LLMs) scale up, MoE has emerged as an effective method to substantially increase model capacity while keeping computational costs manageable. The core idea is that different "experts" specialize in different tasks or data aspects, with only relevant experts activated for a given input.

Background and Fundamentals

An MoE layer typically replaces the feed-forward network (FFN) in transformer blocks and consists of multiple expert networks alongside a gating network. The gating network routes inputs to appropriate experts. The survey distinguishes two main types: - **Dense MoE**: Activates all experts (higher accuracy but computationally expensive) - **Sparse MoE**: Activates only a selected subset (top-k experts), achieving sparsity with reduced computation but requiring load balancing mechanisms to prevent uneven expert utilization

Proposed Taxonomy

The authors introduce a novel taxonomy organizing MoE research into three dimensions: **algorithm design**, **system design**, and **applications**.

Algorithm Design

Gating Functions are categorized into three types: - **Sparse gating** (token-choice, expert-choice, non-trainable variants) - **Dense gating** (effective for LoRA-MoE fine-tuning) - **Soft gating** (token merging and expert merging for full differentiability)

Auxiliary loss functions are critical for load balancing across experts.

Experts: Networks can be FFNs (most common), attention mechanisms, or CNNs. Key hyperparameters include expert count, expert size, and placement frequency. Shared experts have become a mainstream design.

Mixture of Parameter-Efficient Experts (MoPE): Combines MoE with parameter-efficient fine-tuning (PEFT) techniques like LoRA, applied to FFNs, attention, transformer blocks, or every layer.

Training/Inference Schemes: Include dense-to-sparse, sparse-to-dense transitions, and expert model merging approaches.

System Design

The survey addresses challenges from MoE's sparse, dynamic computational workload across three areas: - **Computation**: Load imbalance and specialized GPU kernels (e.g., MegaBlocks, ScatterMoE) - **Communication**: All-to-All communication overhead optimization through hierarchical strategies and computation-communication overlapping - **Storage**: Memory management through selective parameter retention and offloading

Applications

MoE has been successfully applied across: - **Natural Language Processing**: Machine translation, question answering, code generation - **Computer Vision**: Vision MoE (V-MoE), adversarial robustness - **Recommender Systems**: Multi-task learning (MMOE, PLE) - **Multimodal Applications**: Vision-language models (LiMoE, MoE-LLaVA, Uni-MoE)

Challenges and Future Directions

The authors identify key challenges including: training stability and load balancing, scalability and communication overhead, expert specialization and collaboration, sparse activation efficiency, generalization and robustness, interpretability, optimal expert architecture design, and integration with existing frameworks.

Conclusion

This survey serves as an essential reference, providing systematic categorization, open-source implementations, hyperparameter configurations, and empirical evaluations. The authors maintain a resource repository for ongoing updates, aiming to advance MoE research and enable more efficient, powerful machine learning systems capable of handling increasingly complex real-world tasks.